

8.3 Material Calculators

Take-off formulas and coverage tables for the materials you order by the truckload.

How to use these calculators

Use these formulas and calculators to estimate material quantities for your project. Always add 5-15% extra for waste, cuts, and mistakes. Order materials in standard sizes and quantities to minimize cost.

CONCRETE CALCULATOR

$$\text{Cubic Yards} = (\text{Length} \times \text{Width} \times \text{Thickness in inches}) \div 324$$

Example: For a 20' × 30' slab that is 4" thick:
 $(20 \times 30 \times 4) \div 324 = 2,400 \div 324 = 7.4$ cubic yards
 Order 8 cubic yards (rounded up for waste)

Concrete Coverage by Slab Thickness

Slab Thickness	Square Feet per Cubic Yard
4 inches	81 sq ft
5 inches	65 sq ft
6 inches	54 sq ft
8 inches	40 sq ft

DRYWALL CALCULATOR

$$\text{Sheets Needed} = \text{Total Area} \div 32 \text{ (for 4' } \times \text{ 8' sheets)}$$

Example: For a room with 800 sq ft of wall and ceiling area:
 $800 \div 32 = 25$ sheets
 Order 28 sheets (add 10% for waste and cuts)

Drywall Tips:

- Use 4' × 8' sheets for walls, 4' × 12' for ceilings if possible
- 1 gallon joint compound per 100 sq ft of drywall
- 1 roll of tape per 100 linear feet of seams
- 1 pound of screws per 8 sheets (1,000 screws approx)

PAINT CALCULATOR

$$\text{Gallons Needed} = \text{Total Area} \div 350 \text{ (per coat)}$$

Example: For 1,200 sq ft of wall area with 2 coats:

$$1,200 \div 350 = 3.4 \text{ gallons per coat}$$

$$3.4 \times 2 \text{ coats} = 6.8 \text{ gallons total}$$

Order 7 gallons (round up)

Paint Coverage by Surface

Surface Type	Coverage per Gallon
Smooth drywall/plaster	400 sq ft
Textured drywall	350 sq ft
Rough wood/concrete	300 sq ft
Primer on new drywall	300 sq ft

FLOORING CALCULATOR

$$\text{Square Feet Needed} = (\text{Length} \times \text{Width}) \times 1.10 \text{ (waste factor)}$$

Example: For a 12' × 15' room:

$$12 \times 15 = 180 \text{ sq ft}$$

$$180 \times 1.10 = 198 \text{ sq ft}$$

Order 200 sq ft (round up to nearest package size)

Flooring Waste Factors

Flooring Type	Recommended Waste Factor
Carpet, sheet vinyl	5-10% (1.05 - 1.10)
Hardwood, straight pattern	10-15% (1.10 - 1.15)
Tile, diagonal pattern	15-20% (1.15 - 1.20)
Laminate, complex room	10-15% (1.10 - 1.15)

SIDING CALCULATOR

$$\text{Square Feet} = (\text{Total Wall Area}) - (\text{Window/Door Areas})$$

Example: For a wall 40' long × 10' high with two 3' × 5' windows:

Wall area: $40 \times 10 = 400$ sq ft

Window area: $2 \times (3 \times 5) = 30$ sq ft

Net siding: $400 - 30 = 370$ sq ft

With 10% waste: $370 \times 1.10 = 407$ sq ft

Siding Material Tips:

- Add 10-15% waste for lap siding
- Add 15-20% waste for diagonal or fancy patterns
- Vinyl siding sold by "square" (100 sq ft)
- Account for starter strips, J-channels, and trim

ROOFING CALCULATOR

$$\text{Squares Needed} = (\text{Roof Area in sq ft}) \div 100$$

Example: For a roof with 2,400 sq ft of area:

$2,400 \div 100 = 24$ squares

With 10% waste: $24 \times 1.10 = 26.4$ squares

Order 27 squares

Roof Pitch Multipliers

Roof Pitch	Multiplier	Example: 1,000 sq ft floor
Flat to 3:12	1.03	1,030 sq ft roof
4:12	1.05	1,050 sq ft roof
5:12	1.08	1,080 sq ft roof
6:12	1.12	1,120 sq ft roof
8:12	1.20	1,200 sq ft roof
10:12	1.30	1,300 sq ft roof
12:12	1.41	1,410 sq ft roof

Roofing Additional Materials:

- Ridge cap: 1 bundle per 30 linear feet
- Starter strip: 1 bundle per 100 linear feet
- Ice & water shield: 1 roll covers ~2 squares (200 sq ft; 36in × 66.7ft roll)
- Underlayment: 4 squares per roll (400 sq ft coverage)

FRAMING LUMBER ESTIMATOR**Stud, Joist and Plate Counts**

Application	Spacing	Formula
Wall studs	16" O.C.	(Linear feet × 0.75) + extras for corners/openings
Wall studs	24" O.C.	(Linear feet × 0.50) + extras for corners/openings
Floor/ceiling joists	16" O.C.	(Linear feet × 0.75) + 1
Floor/ceiling joists	24" O.C.	(Linear feet × 0.50) + 1
Plates (top/bottom)	Per wall	Linear feet × 3 (for top double plate + bottom)

⚠ IMPORTANT

These calculators provide estimates only. Always verify measurements on-site and consult with suppliers for material recommendations. Order extra materials for waste, mistakes, and future repairs. Keep leftover materials labeled and stored properly for warranty and maintenance purposes.